

Notes: Keister & Sanches (RES, 2023)

Should Central Banks Issue Digital Currency?

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1 Big picture

- **Question.** Should a central bank issue a central bank digital currency (CBDC), and how should its design affect welfare, bank deposits, bank lending, interest rates, and investment?
- **Core mechanism.** A CBDC can improve exchange efficiency by giving households a better liquid asset for transactions. But if it competes with bank deposits, it may reduce the liquidity premium on deposits, raise banks' funding costs, shrink deposits, and reduce bank-financed investment.
- **Main tradeoff.** CBDC improves *decentralized-market exchange* by increasing spendable balances, but it may worsen *centralized-market investment* if deposits fund productive projects and those projects are credit constrained.
- **Key modeling innovation.** The paper embeds both public money and private bank deposits in a Lagos–Wright/New Monetarist environment with credit-constrained bankers. Bank deposits are privately issued inside money that can carry a liquidity premium because they are useful in transactions.
- **Targeted CBDC result.** If the central bank can issue targeted CBDCs, a cash-like CBDC competes only with physical currency, while a deposit-like CBDC competes only with bank deposits. These two designs solve different policy problems and generally require different optimal interest rates.
- **Cash-like CBDC result.** A cash-like CBDC is welfare improving only if socially valuable cash-type transactions are sufficiently important. It has no direct effect on bank funding or investment because it substitutes one form of outside money for another.
- **Deposit-like CBDC result.** A deposit-like CBDC can be welfare improving even though it causes some bank disintermediation. It is especially attractive when productive projects are scarce or when the benchmark economy overinvests because the liquidity premium pushes deposits too far.
- **Universal CBDC result.** If the CBDC cannot be targeted and is accepted in both cash-type and deposit-type transactions, the central bank has only one interest-rate instrument. A universal CBDC is still often desirable, but its use may circulate more narrowly or more widely than a system of targeted CBDCs.
- **Central bank lending result.** Lending to banks can make CBDC holdings positive without changing the real allocation, consistent with the public/private money equivalence idea. But this does not eliminate the fundamental tradeoff when the optimal CBDC policy deliberately increases outside money balances.
- **Policy takeaway.** CBDC design is not merely a technological choice. Whether the CBDC is cash-like, deposit-like, universal, interest-bearing, or accompanied by central bank lending determines whether it improves financial inclusion, crowds out banks, raises or lowers investment, and raises welfare.

2 Incremental contribution of the paper

2.1 What the paper adds relative to policy debate

Policy discussions often frame CBDC as either obviously beneficial because it gives consumers access to safer and more competitive payment services, or dangerous because it disintermediates banks. This paper contributes by showing that both views are incomplete. The right welfare conclusion depends on the *transactional role* of the CBDC and on whether bank deposits create a valuable liquidity premium that helps fund productive investment.

Incremental contribution. The paper gives a formal welfare framework for a policy debate that was often conducted in qualitative terms. It does not simply ask whether deposits shrink. It asks whether the deposits that shrink were financing socially efficient investment and whether the exchange gains from CBDC justify that loss.

2.2 What the paper adds relative to New Monetarist models

The model builds on Lagos–Wright style monetary theory, but adds a key role for *bank deposits* as privately issued money. Deposits are useful for exchange, and this use can lower banks’ funding costs. At the same time, bankers face a pledgeability constraint, so lower funding costs can affect how much investment is undertaken.

Incremental contribution. The paper connects the liquidity premium on deposits to bank intermediation and CBDC design. It shows that introducing public digital money changes both the payment system and the real allocation of capital.

2.3 What the paper adds relative to prior CBDC theory

Compared with models in which CBDC mainly disciplines bank market power or expands central-bank balance sheets, Keister and Sanches emphasize a competitive banking environment with credit frictions. Their central point is that a CBDC may reduce a liquidity premium that is either socially helpful or socially harmful depending on the investment margin.

Incremental contribution. The paper identifies conditions under which disintermediation is welfare improving and conditions under which it is costly. This makes the paper useful for central-bank design choices: the same CBDC interest rate can be good or bad depending on scarcity of productive projects, pledgeability, and the type of transactions served.

2.4 What the paper adds through targeted versus universal CBDC

A major contribution is the separation between *targeted* and *universal* CBDC. Targeted CBDCs can be designed to compete with one existing means of payment. Universal CBDC competes with both cash and deposits, creating interactions across margins.

Incremental contribution. The paper shows that technological design restrictions are economically important. If the central bank can target the CBDC, it can use separate interest rates to manage separate tradeoffs. If the CBDC is universal, the same interest rate must balance financial inclusion, illicit-transaction concerns, efficient exchange, bank funding, and investment.

3 Data, measurement, and model primitives

This is a theoretical paper, so the relevant “data and measurement” objects are the primitives and state variables of the model rather than an empirical dataset.

3.1 Markets and timing

Each period has two subperiods.

- In the **centralized market** (CM), agents can trade goods and rebalance portfolios.
- In the **decentralized market** (DM), buyers and sellers meet bilaterally and money is essential because identities and histories are private.
- Buyers make portfolio choices before knowing whether they will be in a transaction where cash, deposits, or CBDC can be recognized.

Interpretation. The CM is where investment and portfolio adjustment happen. The DM is where the means-of-payment role of assets matters. CBDC affects the economy because it changes which assets are useful in DM exchange and what return those assets carry.

3.2 Agents

The economy has three types of agents.

- **Buyers** want to consume the DM good but cannot produce it.
- **Sellers** can produce the DM good but do not consume it.
- **Bankers** have projects that require one unit of CM input and return γ_j units of CM output in the next period.

Bankers are heterogeneous. There is a measure η of bankers at each productivity level $\gamma \in [0, \bar{\gamma}]$. The parameter η therefore controls the density or abundance of investment opportunities.

Interpretation. When η is large, productive projects are plentiful and crowding out bank deposits is more costly. When η is small, projects are scarce, and reducing bank-funded investment can be less costly or even beneficial if the economy overinvests.

3.3 Payment technologies

There are two types of DM meetings.

- **Type 1 meetings:** sellers can verify physical currency but not bank deposits. These represent low-value, privacy-sensitive, offline, or cash-style transactions.
- **Type 2 meetings:** sellers can verify bank deposits but not physical currency. These represent transactions where electronic bank-based payment is useful, such as large-value or remote trades.
- CBDC design determines whether digital currency can be verified by Type 1 sellers, Type 2 sellers, or both.

Interpretation. This verification structure is the central design margin. A cash-like CBDC is accepted by Type 1 sellers. A deposit-like CBDC is accepted by Type 2 sellers. A universal CBDC is accepted by both.

3.4 Welfare and social value of cash-type transactions

The paper allows the planner to discount the social value of Type 1 transactions by a parameter $\nu \in [0, 1]$. This may capture illicit activity or other transactions that are privately useful but not fully socially valuable.

Aggregate welfare is

$$\sum_{t=0}^{\infty} \beta^t [X_t^b + X_t^s + X_t + \alpha \{ \lambda_1 (\nu u(q_{1t}) - w(q_{1t})) + \lambda_2 (u(q_{2t}) - w(q_{2t})) \}]. \quad (1)$$

Interpretation. The parameter ν is the policy weight on cash-like transactions. A high ν means promoting financial inclusion or cash-style payment efficiency is important. A low ν means the central bank worries that making cash-like transactions easier may facilitate socially undesirable activity.

The value function $V_s(\mathbf{a}', t)$ satisfies

$$V_s(\mathbf{a}', t) = \alpha [u(q_s(\mathbf{a}', t)) + \beta J(\mathbf{a}' - \mathbf{h}_s(\mathbf{a}', t), t+1)] + (1-\alpha)\beta J(\mathbf{a}', t+1),$$

where $q_s(\mathbf{a}', t) \in \mathbb{R}_+$ denotes the buyer's consumption of the DM good and $\mathbf{h}_s(\mathbf{a}', t) \in \mathbb{R}_+^3$ denotes the payment she makes for this consumption out of her asset holdings $\mathbf{a}'$. The function $J(\mathbf{a}, t)$ in this expression is the expected value of entering the CM before knowing the type of her potential decentralized meeting in the following period,

$$J(\mathbf{a}, t) \equiv \lambda_1 J_1(\mathbf{a}, t) + \lambda_2 J_2(\mathbf{a}, t).$$

Bargaining. Throughout the analysis, we assume that buyers make a take-it-or-leave-it offer to sellers, solving

$$\max_{(q_s, \mathbf{h}_s) \in \mathbb{R}_+^4} [u(q_s) - \beta \mathbf{R}_t \cdot \mathbf{h}_s]$$

subject to the seller's participation constraint

$$-w(q_s) + \beta \mathbf{R}_t \cdot \mathbf{h}_s \geq 0$$

and the liquidity constraint

$$\mathbf{h}_s \leq \mathbf{f}_s(\mathbf{a}). \quad (3)$$

The function $\mathbf{f}_s$ requires the buyer to pay with assets her trading partner can verify. If, for example, Type 1 sellers can only verify physical currency, we have $\mathbf{f}_1(\mathbf{a}) = (m, 0, 0)$. If these sellers can instead verify both physical and digital currency, we have $\mathbf{f}_1(\mathbf{a}) = (m, 0, e)$. In the sections that follow, we impose particular functions $\mathbf{f}_s$ to capture different potential digital currency designs. For now, we only impose that Type 1 sellers can verify physical currency but not bank deposits and that the reverse holds for Type 2 sellers.

The solution to this problem implies the following schedule for DM output

$$q_s(\mathbf{a}, t) = \begin{cases} w^{-1}(\beta \mathbf{R}_t \cdot \mathbf{f}_s(\mathbf{a})) & \text{if } \mathbf{R}_t \cdot \mathbf{f}_s(\mathbf{a}) < \frac{w(q^*)}{\beta} \\ q^* & \text{otherwise} \end{cases} \quad (4)$$

and for payments

$$\mathbf{f}_s(\mathbf{a}, t) = \begin{cases} \mathbf{f}_s(\mathbf{a}) & \text{if } \mathbf{R}_t \cdot \mathbf{f}_s(\mathbf{a}) < \frac{w(q^*)}{\beta} \end{cases}$$

Original excerpt: bargaining and liquidity constraint

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Aggregate welfare is then

$$\sum_{t=0}^{\infty} \beta^t \left[X_t^b + X_t^s + X_t + \alpha \left[\lambda_1 (vu(q_{1t}) - w(q_{1t})) + \lambda_2 (u(q_{2t}) - w(q_{2t})) \right] \right]. \quad (1)$$

The feasibility of an allocation requires the net consumption of all agents in the CM to be no greater than the net output of bankers' investment projects. We focus on allocations characterized by a cutoff value $\hat{\gamma}_t$ above which a banker's project is operated and below which it is not. Feasibility in period t then requires

$$X_t^b + X_t^s + X_t \leq \eta \int_{\hat{\gamma}_t}^{\bar{\gamma}} \gamma d\gamma - \eta(\bar{\gamma} - \hat{\gamma}_t). \quad (2)$$

The right-hand side of this expression is the output from projects maturing at the current date

Original excerpt: welfare and feasibility conditions

3.5 Assets and returns

A buyer can hold a portfolio

$$a = (m, d, e),$$

where m is physical currency, d is bank deposits, and e is CBDC. The real return on physical currency is determined by inflation. Deposits pay real interest $1 + r_t$. CBDC pays nominal interest $1 + i$ and therefore has a real return depending on i and inflation.

Interpretation. CBDC is not just another medium of payment. If it bears interest, it becomes a new policy instrument that changes buyers' portfolio choices and banks' funding conditions.

3.6 Deposit supply and pledgeability

A banker can issue deposits only if promised repayment does not exceed pledgeable project income:

$$1 + r_t \leq \theta \gamma_j. \quad (2)$$

The cutoff banker has

$$\hat{\gamma}_t = \frac{1 + r_t}{\theta}. \quad (3)$$

The aggregate supply of deposits is therefore increasing when the real deposit rate falls because more bankers satisfy the borrowing constraint.

Interpretation. This pledgeability constraint links payment-system competition to real investment. If CBDC raises the return households require on liquid assets, banks must pay more to attract deposits. Higher deposit rates raise the cutoff $\hat{\gamma}$, so fewer projects are funded.

only on whether the seller accepts the asset but also on whether the buyer is liquidity constrained. Define the function $L: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ by

$$L(A) = \begin{cases} \alpha \frac{w^{-1}(\beta A)}{w^{-1}(\beta A)} + 1 - \alpha & \text{if } A \leq w(q^*)/\beta \\ 1 & \text{otherwise.} \end{cases}$$

This function measures the expected benefit of holding an extra unit of spendable assets. If the buyer's current spendable assets are insufficient to purchase the efficient quantity q^* , the increase will allow her to consume more if she is matched in the DM, which occurs with probability α . If she is not matched, or if she already has enough spendable assets to purchase q^* , she merely holds the extra unit of assets until the following CM.

Using this function, the first-order condition for the real physical currency balances of a buyer who will potentially be in a Type 1 match can be written as

$$L(\mathbf{R}_t \cdot \mathbf{f}_1(\mathbf{a}')) \leq \frac{\phi_t}{\beta \phi_{t+1}}, \quad (5)$$

with equality if $m' > 0$. The first-order condition for the deposits of a buyer who will potentially be in a Type 2 match is

$$L(\mathbf{R}_t \cdot \mathbf{f}_2(\mathbf{a}')) \leq \frac{1}{\beta(1+r_t)}, \quad (6)$$

with equality if $d' > 0$. In addition, only buyers potentially entering Type 1 meetings will hold physical currency and only buyers potentially entering Type 2 meetings will hold bank deposits.⁸ If Type s sellers accept digital currency, the first-order condition for real digital currency balances of a buyer who will potentially be in a Type s match is

$$L(\mathbf{R}_t \cdot \mathbf{f}_s(\mathbf{a}')) \leq \frac{\phi_t}{\beta(1+i)\phi_{t+1}}, \quad (7)$$

with equality if $e' > 0$. Equations (5)–(7) thus characterize the demand for each asset in the period- t CM.

2.3. Asset supply

Deposits. To derive the supply of deposits, consider a banker born in period t with a project that returns $y_t \in [0, \bar{y}]$. Given a market interest rate r_t , this banker is willing to issue a deposit and invest if

$$y_t - (1+r_t) \geq 0.$$

However, the promised repayment on this deposit cannot exceed the value of the banker's pledgeable future income

$$1 + r_t < \theta v_t. \quad (8)$$

Original excerpt: asset demand and pledgeability

Let $\hat{y}_t \in \mathbb{R}_+$ denote the banker whose project's payoff satisfies the pledgeability restriction with equality in period t , that is,

$$\hat{y}_t = \frac{1+r_t}{\theta}. \quad (9)$$

The aggregate supply of deposits then equals the measure of bankers with project returns of at least $\hat{y}_t$, which equals $\eta(\bar{y} - \hat{y}_t)$, or

$$\eta \left(\bar{y} - \frac{1+r_t}{\theta} \right).$$

Note that, for any θ , a reduction in the interest rate leads to an increase in investment by allowing a larger number of bankers to issue deposits. In other words, a lower interest rate will lead to an expansion of the banking system and an increased supply of deposits.

Currency. The supply of both physical and digital currency is set by the central bank following a price-level target rule with

$$\frac{\phi_t}{\phi_{t+1}} = \mu > \beta \quad \text{for all } t.$$

The central bank stands ready to exchange units of both physical and digital currency for CM goods at the desired price level each period.⁹ In addition to pinning down the price level, this policy ensures that digital and physical currency always have the same value.¹⁰ Letting $\bar{M}_t \in \mathbb{R}_+$ denote the supply of physical currency and $\bar{E}_t$ the supply of digital currency, the central bank's budget constraint is

$$\phi_t(\bar{M}_t + \bar{E}_t) = \phi_t(\bar{M}_{t-1} + (1+i)\bar{E}_{t-1}) + \tau_t,$$

where the lump-sum tax/transfer τ_t is set to balance the budget each period.¹¹

2.4. Market clearing

Because physical currency is only used in Type 1 meetings, we can write its market clearing equation as

$$\lambda_1 m_t = \phi_t \bar{M}_t. \quad (10)$$

Similarly, the fact that bank deposits are only exchanged in Type 2 meetings allows us to write the market-clearing equation for the deposit market as

$$\lambda_2 d_t = \eta \left(\bar{y} - \frac{1+r_t}{\theta} \right). \quad (11)$$

Market clearing for digital currency requires

–

Original excerpt: deposit supply and market clearing

4 Benchmark economy without CBDC

4.1 Equilibrium allocation

Without CBDC, Type 1 buyers use physical currency and Type 2 buyers use bank deposits:

$$f_1(a) = (m, 0, 0), \quad f_2(a) = (0, d, 0).$$

The equilibrium is stationary and characterized by asset-demand conditions and market-clearing equations for currency and deposits. The deposit rate $1 + r^N$ is the key endogenous object.

4.2 Liquidity premium on deposits

Deposits may carry a liquidity premium because they are useful in Type 2 transactions. If households value their use in exchange, banks can fund projects at a lower rate than the standard time-preference rate β^{-1} .

- No liquidity premium: $1 + r^N = \beta^{-1}$.
- Liquidity premium: $1 + r^N < \beta^{-1}$.

Interpretation. The liquidity premium can be socially useful or socially harmful. It is useful if it allows more efficient projects to be financed. It is harmful if it pushes investment beyond the efficient level.

4.3 Overinvestment and underinvestment

If $\theta = 1$, a liquidity premium tends to generate overinvestment because any project with return below β^{-1} is inefficient but may be funded due to cheap deposits. If $\theta < 1$, the pledgeability constraint can make investment too low. In that case, the liquidity premium on deposits may offset the friction and move investment closer to efficient.

Interpretation. This is the foundational welfare ambiguity. A CBDC that reduces the deposit liquidity premium can improve welfare if the baseline overinvests, but can reduce welfare if the liquidity premium was helping overcome credit frictions.

4.4 Figure 1: equilibrium deposit rate

Read:

- The blue line is the equilibrium gross deposit rate $1 + r^N$ as a function of pledgeability θ .
- The dashed red line marks θ/β , which corresponds to the boundary for whether the marginal project is efficiently funded.
- The black dashed line is β^{-1} , the return without a liquidity premium.
- When $1 + r^N < \beta^{-1}$, deposits have a liquidity premium.
- When $1 + r^N$ lies below or above θ/β , investment can be inefficiently high or low.

Economic message: The figure shows that the same liquidity premium can have opposite welfare implications depending on pledgeability and project scarcity. This makes the CBDC policy problem nontrivial.

3. EQUILIBRIUM WITH NO DIGITAL CURRENCY

When there is no digital currency, the functions $\mathbf{f}_i$ in the buyer's liquidity constraint in equation (3) are given by

$$\mathbf{f}_1(\mathbf{a}) = (m, 0, 0) \quad \text{and} \quad \mathbf{f}_2(\mathbf{a}) = (0, d, 0).$$

A buyer can only use her physical currency balances in a Type 1 meeting and her bank deposits in a Type 2 meeting. The Inada condition on buyers' utility function then implies that the first-order conditions (5) and (6) for buyers' portfolio choices will hold with equality. Combining these equations with the market-clearing conditions (10) and (11) yields

$$L\left(\frac{m}{\mu}\right) = \frac{\mu}{\beta} \quad (13)$$

and

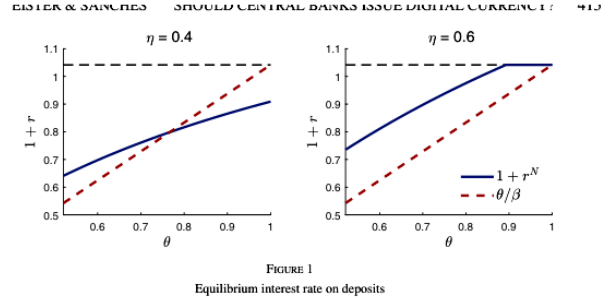
$$L\left((1+r)\frac{\eta}{\lambda_2}\left(\tilde{y} - \frac{1+r_i}{\theta}\right)\right) = \frac{1}{\beta(1+r)}. \quad (14)$$

The fact that only period- t variables appear in each of these two equations shows that an equilibrium is necessarily stationary. To guarantee the existence and uniqueness of equilibrium, we assume preferences are such that:

(i) $AL(A)$ is strictly increasing and (ii) $\lim_{(1+r) \rightarrow 0} \frac{L^{-1}\left(\frac{1}{\beta(1+r)}\right)}{1+r} < \eta\tilde{y}$.

The first assumption ensures the demand for deposits is strictly increasing in the interest rate, while the second guarantees the supply of deposits is large enough to meet demand at some interest rate.¹² The following proposition establishes the existence and uniqueness of equilibrium in the benchmark economy. Proofs of all propositions are contained in the [Supplementary Appendix](#).

Proposition 1. *The economy with no digital currency has a unique equilibrium. There is a liquidity premium on deposits in this equilibrium if and only if $\eta < \frac{\lambda_2 w(q)}{\tilde{y} - \beta_0}$. When this condition holds, the equilibrium interest rate is a strictly increasing function of η .*



Original excerpt: benchmark equilibrium and Proposition 1

Figure 1: equilibrium interest rate on deposits

5 Targeted digital currencies

The paper first studies CBDCs that can be targeted to one type of transaction. This is the clean benchmark because the central bank can decide whether the CBDC competes with cash or deposits.

5.1 Cash-like CBDC

A cash-like CBDC can be verified only in Type 1 meetings. It competes with physical currency but not with bank deposits:

$$f_1(a) = (m, 0, e), \quad f_2(a) = (0, d, 0).$$

5.1.1 Equilibrium effect

If the cash-like CBDC pays no interest, it is just another form of outside money and leaves real allocations unchanged. If it pays a positive interest rate or provides nonpecuniary benefits, buyers hold more spendable outside money in Type 1 meetings. This raises Type 1 DM output.

Interpretation. Because the cash-like CBDC cannot be used in deposit-type meetings, it does not crowd out bank deposits. The policy tradeoff is not about intermediation; it is about the social value of cash-like transactions.

5.1.2 Optimal policy

The cash-like optimal policy balances two forces.

- Higher CBDC return raises the quantity traded in Type 1 meetings.
- The social planner values those trades only by ν , not necessarily by full private surplus.

The optimal gross interest rate is

$$1 + i^C = \frac{\mu}{\beta} \frac{\nu}{\alpha + (1 - \alpha)\nu} \quad (4)$$

when the cash-like CBDC is desirable.

Interpretation. This is a modified Friedman rule. If $\nu = 1$, the planner wants to make cash-like exchange efficient. If ν is low, the planner suppresses cash-like digital currency because it may facilitate socially undesirable transactions.

between the two currencies and the equilibrium paths of $\{m_t\}$ and $\{e_{1t}\}$ are indeterminate, but total real money balances and all other equilibrium quantities are unchanged. Therefore, introducing a cash-like CBDC will only affect equilibrium consumption and welfare if it carries a positive interest rate.¹⁶

When $i > 0$, the first-order condition in equation (7) will hold with equality and the equilibrium quantity of real currency balances e_{1t} will satisfy

$$L\left(\frac{(1+i)}{\mu}e_{1t}\right) = \frac{\mu}{(1+i)\beta}.$$

Comparing this expression with equation (13) shows that $i > 0$ implies buyers entering a Type 1 meeting will hold larger real currency balances when the digital currency is introduced. Equation (4) then shows that the quantity of DM production in Type 1 meetings also increases. Combining the different cases, we can express this quantity as

$$q_1^C(1+i) = w^{-1}\left(\beta L^{-1}\left(\frac{\mu}{\beta \max\{1+i, 1\}}\right)\right). \quad (15)$$

The equilibrium quantities of deposits, CM investment, and DM production in Type 2 meetings are unchanged. The following proposition summarizes these results, using a superscript C to denote equilibrium values with a cash-like CBDC.

Proposition 2. *Under a cash-like digital currency with $i > 0$, the unique equilibrium allocation satisfies $e_1^C > m^N$, $q_1^C > q_1^N$ and $(q_2^C, \hat{y}^C) = (q_2^N, \hat{y}^N)$.*

Optimal policy. The potential benefit of introducing a cash-like CBDC in our framework is that it allows the policymaker to increase the rate of return on currency while maintaining the same target for the price level and inflation. We think of this target as being determined by considerations outside the scope of our model, such as stabilization policy and the zero lower bound. (See Andrade, Gali, Le Bihan and Matheron (2019) for a recent discussion.) If the policymaker could

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Proposition 3. *There exists $\bar{\nu} \in (0, 1)$ such that a cash-like digital currency is desirable if and only if $\nu > \bar{\nu}$. In this case, the optimal policy is given by*

$$1 + i^C = \frac{\mu}{\beta} \frac{\nu}{\alpha + (1 - \alpha)\nu}. \quad (16)$$

When ν is large, the policymaker wants cash buyers to have access to a better means of payment, which we interpret as a desire to promote *financial inclusion*. When ν is small, in contrast, concerns about facilitating illicit activities make a cash-like CBDC undesirable. The interest rate

Original excerpt: cash-like CBDC allocation and Proposition 2

Original excerpt: Proposition 3 cash-like desirability

5.2 Deposit-like CBDC

A deposit-like CBDC can be verified only in Type 2 meetings. It competes with bank deposits but not with physical currency:

$$f_1(a) = (m, 0, 0), \quad f_2(a) = (0, d, e).$$

5.2.1 When is a deposit-like CBDC held?

Type 2 buyers hold whichever liquid asset has the higher return. A deposit-like CBDC affects equilibrium only if

$$1 + i > \mu(1 + r^N), \quad (5)$$

where $\mu(1 + r^N)$ is the nominal return on deposits in the no-CBDC economy.

Interpretation. The central bank must make the CBDC attractive enough to compete with bank deposits. If it is not attractive enough, households hold deposits and the CBDC is irrelevant.

5.2.2 Disintermediation mechanism

Once the deposit-like CBDC is attractive, Type 2 buyers shift some balances from deposits to CBDC. The supply of bank deposits falls. Banks' funding costs rise. The cutoff $\hat{\gamma}$ rises. Some bankers whose projects were previously funded no longer invest.

Interpretation. Disintermediation is not an incidental feature of the policy. In this model, the welfare benefit from deposit-like CBDC comes from changing the total quantity and composition of liquid assets. That necessarily interacts with bank deposit creation.

5.2.3 Marginal cost–benefit tradeoff

Raising the CBDC interest rate has:

- a **marginal cost**: fewer bank projects are funded, reducing CM output;
- a **marginal benefit**: Type 2 buyers have more real balances, raising DM exchange efficiency.

Interpretation. The CBDC interest rate becomes a policy lever that trades off efficient exchange against efficient investment.

Equilibrium. With a deposit-like CBDC, the functions $\mathbf{f}_i$ in the buyer's liquidity constraint in equation (3) become

$$\mathbf{f}_1(\mathbf{a}) = (m, 0, 0) \quad \text{and} \quad \mathbf{f}_2(\mathbf{a}) = (0, d, e).$$

Buyers in a Type 1 meeting can only use their physical currency balances to make purchases, while buyers in a Type 2 meeting can use their bank deposits and/or digital currency balances. Comparing the first-order conditions for bank deposits and digital currency in equations (6) and (7) shows that a Type 2 buyer would only choose to hold whichever asset offers the higher return. If the nominal interest rate on the digital currency, $1 + i$, is set below the equilibrium nominal interest rate on deposits in the benchmark model, $\mu(1 + r^N)$, the demand for the digital currency will be zero, and it will have no effect on the equilibrium allocation. If these two interest rates are equal, Type 2 buyers would be indifferent between deposits and digital currency. However, the first-order condition for deposits (6) together with the market-clearing condition (11) would imply that equilibrium digital currency holdings must again be zero. A deposit-like CBDC will only be held in equilibrium if the interest rate is set higher than the nominal interest rate on deposits in the benchmark model, that is, if

$$1 + i > \mu(1 + r^N). \quad (17)$$

Note that this condition can be satisfied only if there is a liquidity premium on deposits in the benchmark model, that is, if $1 + r^N < \beta^{-1}$. In other words, the assets that back bank deposits must be in excess supply for a deposit-like CBDC to play a role in our model.

Original excerpt: deposit-like CBDC condition

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$$1 + i = \mu(1 + r^D), \quad (18)$$

where the superscript D denotes equilibrium values under a deposit-like regime. Because individual Type 2 buyers are indifferent between the two assets in equilibrium, the first-order conditions (6) and (7) will both hold with equality and a Type 2 buyer's total holding of spendable assets will satisfy

$$L\left((1 + r^D)\left[\frac{\eta}{\lambda_2}\left(\tilde{\gamma} - \frac{1 + r^D}{\theta}\right) + e_2^D\right]\right) = \frac{1}{\beta(1 + r^D)}. \quad (19)$$

The higher interest rate on deposits implies through equations (9) and (11) that the level of CM investment and the quantity of deposits issued by banks both decrease. In other words, a deposit-like CBDC disintermediates banks to some degree and crowds out bank-financed investment. At the same time, however, equation (19) shows that the higher rate of return leads Type 2 buyers to hold larger total spendable assets, $e_2 + d$. These larger asset holdings translate, through equation (6), into larger production of the Type 2 good in DM meetings. Finally, because the digital currency cannot be used in Type 1 DM meetings, the demand for physical currency and the quantity produced in Type 1 meetings are unchanged from the benchmark case. The following proposition summarizes these results.

Proposition 4. With a deposit-like digital currency satisfying condition (17), the unique equilibrium allocation satisfies $e_2^D + d^D > e_2^N > d^D$, $(r^D, \tilde{\gamma}^D, q_2^D) \gg (r^N, \tilde{\gamma}^N, q_2^N)$, and $q_1^D = q_1^N$.

Optimal policy. The results in Proposition 4 point to a tradeoff the central bank faces when setting the interest rate on a deposit-like CBDC. Raising this interest rate increases DM output in Type 2 meetings and promotes efficient exchange, but decreases CM investment. The optimal policy balances these competing concerns. Using the stationarity of the equilibrium allocation and omitting the terms that are unaffected by a deposit-like CBDC, we can write the objective in equation (1) as

$$r\tilde{\gamma}$$

Original excerpt: Proposition 4 deposit-like allocation

5.3 Desirability of deposit-like CBDC

Proposition 5 shows that a deposit-like CBDC is desirable when productive projects are sufficiently scarce. Corollary 1 gives a particularly transparent case: if the benchmark economy overinvests, a deposit-like CBDC improves welfare by crowding out inefficient projects and increasing the Type 2 DM exchange quantity.

Interpretation. The result directly responds to policy fears about disintermediation. The paper does not deny that CBDC can disintermediate banks. It shows that disintermediation can be optimal if the displaced bank funding was supporting projects with low social value.

5.4 Figure 2: optimal deposit-like CBDC policy

Read:

- The left panel compares the benchmark deposit rate and the optimal deposit-like CBDC rate.
- The optimal CBDC rate lies above the benchmark deposit rate, so the CBDC actively competes with deposits.
- As pledgeability θ rises, the optimal deposit-like CBDC rate approaches the Friedman-rule upper bound.
- The right panel shows the welfare gain is hump-shaped in θ .

Economic message: Welfare gains are largest when financial frictions are neither too severe nor too weak. If frictions are too severe, crowding out bank finance is costly. If frictions are too weak, the private system already works relatively well.

Proposition 5. *There exists $\bar{\eta} > 0$ such that $\eta < \bar{\eta}$ implies a deposit-like digital currency is desirable. The optimal policy satisfies $1 + i^D \in [\mu(1 + r^N), \mu/\beta]$ if $\theta < 1$ and $1 + i^D = \mu/\beta$ if $\theta = 1$.*

One case where it is easy to see that a deposit-like CBDC can raise welfare is when the equilibrium without a digital currency exhibits overinvestment. If $1 + r^N < \theta/\beta$ holds, the marginal project being funded in the economy without digital currency returns less than β^{-1} . Introducing a CBDC that bears a slightly higher real return will crowd out these inefficient projects while at the same time increasing the quantity produced in Type 2 DM meetings toward the efficient level. We record this result in the following corollary.

Corollary 1. *If $1 + r^N < \frac{\theta}{\beta}$, a deposit-like digital currency is desirable.*

Figure 2 depicts the optimal interest rate on a deposit-like CBDC and the associated welfare gain, using the same parameter values as the left panel in Figure 1. The left panel in Figure 2 shows how, for this example, the optimal real interest rate on a deposit-like digital currency is strictly higher than the interest rate on deposits in the baseline economy for all values of θ . Moreover, the optimal interest rate converges to μ/β as θ approaches 1. The right panel of Figure 2 shows that the welfare gain from introducing a digital currency is largest for intermediate values of the credit friction θ . Two competing forces are at work. On one hand, as θ increases, the costs associated with crowding out CM investment become smaller. In particular, the measure of projects that are crowded out by a marginal increase in the deposit rate and the productivity of the marginal project are both nonincreasing to θ^{-1} . On the other hand, an increase in θ decreases the liquidity

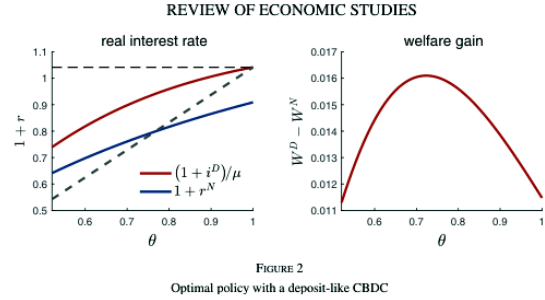


FIGURE 2
Optimal policy with a deposit-like CBDC

Original excerpt: Proposition 5 and Corollary 1

Figure 2: optimal policy with a deposit-like CBDC

5.5 Dual targeted CBDCs

If the central bank can issue both a cash-like CBDC and a deposit-like CBDC, it can set two separate interest rates. One rate manages the Type 1 cash-like transaction margin; the other manages the Type 2 deposit/intermediation margin.

5.6 Figure 3: desirability of targeted CBDCs

Read:

- Figure 3 maps parameter combinations (θ, ν) into regions where cash-like CBDC, deposit-like CBDC, or both are desirable.
- Cash-like CBDC tends to be desirable when ν is high, because the social value of Type 1 transactions is high.
- Deposit-like CBDC tends to be desirable when η is small enough and when the intermediation cost of disintermediation is not too high.
- A dual-CBDC system can be optimal when both margins matter simultaneously.

Economic message: The figure shows that “CBDC” is not one policy. Cash-like and deposit-like designs solve different problems. The case for CBDC depends on which margin the central bank wants to improve.

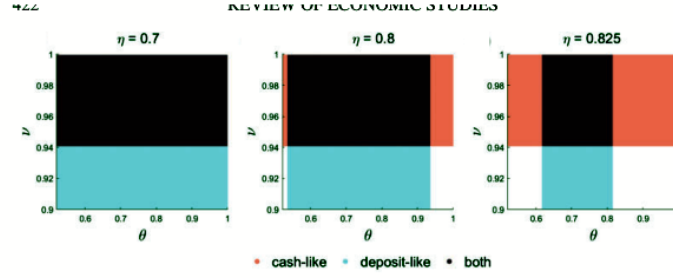


FIGURE 3
Desirability of targeted CBDCs

6 Universal digital currency

6.1 Why universal CBDC is harder

A universal CBDC is accepted by all sellers:

$$f_1(a) = (m, 0, e), \quad f_2(a) = (0, d, e).$$

The central bank has only one interest rate i on the CBDC, and that single instrument affects both Type 1 and Type 2 meetings.

Interpretation. Universal CBDC is more realistic if the central bank cannot prevent the digital currency from being used as a substitute for both cash and deposits. But it creates a policy constraint: the same interest rate must manage multiple margins.

6.2 Equilibrium cases

The paper separates cases by the CBDC interest rate.

- If $i < 0$, Type 1 buyers do not hold CBDC; the policy behaves like deposit-like CBDC.
- If $i > 0$ and $1 + i \leq \mu(1 + r^N)$, Type 1 buyers hold CBDC but Type 2 buyers do not; the policy behaves like cash-like CBDC.
- If $1 + i > \mu(1 + r^N)$, both Type 1 and Type 2 buyers may hold CBDC; the policy combines cash-like and deposit-like effects.

Interpretation. A universal CBDC may be legally or technologically universal, but in equilibrium it may circulate only in one type of transaction if its interest rate is set in the relevant region.

6.3 Universal CBDC objective

The policymaker maximizes welfare over one interest rate. The objective contains three pieces:

- the effect on CM investment through the bank-deposit margin;
- the effect on Type 1 DM exchange;
- the effect on Type 2 DM exchange.

Interpretation. The universal-CBDC problem nests the targeted-CBDC problems but adds an implementation constraint: the same return on CBDC is seen by both types of buyers.

6.4 Figure 4: optimal CBDC interest rates

Read:

- The blue line is the optimal cash-like CBDC rate.
- The red line is the optimal deposit-like CBDC rate.
- The green dashed line is the optimal universal CBDC rate.
- When the targeted optimal rates differ substantially, the universal rate compromises between them.

Economic message: The universal CBDC cannot generally implement the same allocation as two targeted CBDCs. The central bank must sacrifice on one margin to gain on the other.

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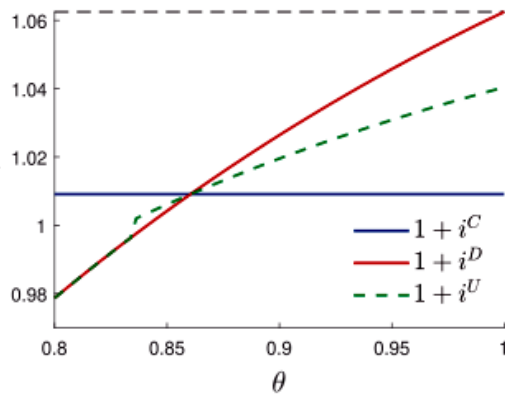


FIGURE 4

Figure 4: optimal CBDC interest rates

Our next result characterizes the conditions under which the constraint associated with a universal CBDC does not bind at the solution to the optimal policy problem.

Proposition 6. *The optimal policy under a universal CBDC implements the same allocation as under two targeted CBDCs if and only if at least one of the following conditions holds:*

- (i) $1 + i^C = 1$ and $1 + i^D = \mu(1 + r^N)$,
- (ii) $1 + i^C = 1 + i^D$,
- (iii) $1 + i^C \leq 1 + i^D = \mu(1 + r^N)$, or
- (iv) $1 + i^C = 1 \geq 1 + i^D$.

It is straightforward to see that each of these four conditions is sufficient to ensure that a universal CBDC can implement the same allocation as two targeted CBDCs under the optimal policy. Condition (i) corresponds to a situation where neither of the targeted CBDCs is desirable. The same allocation can trivially be implemented with a universal CBDC by setting the interest rate low enough that no one chooses to hold it. In condition (ii), the two targeted CBDCs have exactly the same optimal interest rate. Setting the interest rate on a universal CBDC to this common value clearly leads to the same equilibrium allocation. Under condition (iii), a deposit-like CBDC is not desirable and the optimal interest rate on a cash-like CBDC is low enough that no Type 2 buyers would choose to hold it. In this case, the optimal policy sets $i^U = i^C$ and the digital currency is only held by Type 1 buyers. Finally, under condition (iv), a cash-like CBDC is not desirable and the optimal nominal interest rate on the deposit-like CBDC is non-positive. The optimal policy then sets $i^U = i^D < 0$, which ensures the digital currency is only held by Type 2 buyers.

The less obvious part of Proposition 6 is that these four conditions are the *only* situations in which a universal CBDC can implement the optimal allocation with two targeted CBDCs. In all other cases, the restriction that the CBDC interest rate must be the same in both types of

Original excerpt: Proposition 6 universal versus targeted allocation

6.5 When does universal CBDC reproduce targeted outcomes?

Proposition 6 gives conditions under which universal CBDC can implement the same allocation as targeted CBDCs. These cases arise when the desired targeted interest rates are compatible with the same asset-holding pattern in equilibrium.

Interpretation. When the constraints do not bind, universal CBDC is as good as targeted CBDCs. When they bind, the central bank faces a genuine design loss from lacking targeting technology.

6.6 Figure 5: desirability and usage of universal CBDC

Read:

- Figure 5 maps where a universal CBDC is used in Type 1 meetings, Type 2 meetings, or both.
- Relative to Figure 3, universal CBDC can circulate more narrowly because the central bank may choose an interest rate that prevents one group from holding it.
- It can also circulate more widely when avoiding circulation in one type of meeting would require a distorted interest rate.

Economic message: Universal acceptance does not imply universal use. The central bank may deliberately set the interest rate so that only cash users or only deposit users hold CBDC.

6.7 Desirability of universal CBDC

Proposition 7 gives sufficient conditions for universal CBDC to be desirable. The main logic is:

- if Type 1 exchange is socially valuable and the CBDC can improve it without harming Type 2 investment, universal CBDC helps;
- if the benchmark economy overinvests, universal CBDC can improve Type 2 exchange and reduce excessive investment;
- if productive projects are scarce, universal CBDC can be desirable because the opportunity cost of disintermediation is low.

Interpretation. The universal-CBDC result is more nuanced than the targeted result, but the same economic forces remain: financial inclusion and exchange efficiency versus bank intermediation and investment.

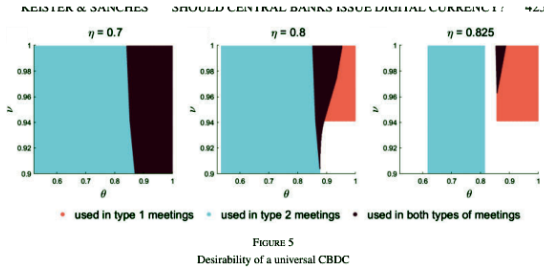


Figure 5: desirability of universal CBDC

Proposition 7. A universal digital currency is desirable if any of the following sets of conditions holds:

$$\begin{aligned}
 & \text{(i) } v > \frac{\alpha\beta}{\mu - (1-\alpha)\beta} \quad \text{and} \quad 1+r^N > \frac{1}{\mu}; \\
 & \text{(ii) } v > \frac{\alpha\beta}{\mu - (1-\alpha)\beta} \quad \text{and} \quad 1+r^N < \frac{\theta}{\beta}; \quad \text{or} \\
 & \text{(iii) } 1+r^N < \frac{1}{\mu} \quad \text{and} \quad 1+r^N < \frac{\theta}{\beta}.
 \end{aligned}$$

In the first two cases, the policymaker would like to increase production and exchange in Type 1 meetings and, therefore, the optimal interest rate on a cash-like CBDC would be positive. With a universal CBDC, however, the policymaker needs to also take into account its effects on Type 2 meetings and CM investment. In case (i), the nominal interest rate on deposits in the baseline economy with no CBDC is strictly positive. The policymaker can, therefore, introduce a digital currency with a small positive interest rate that improves efficiency in Type 1 meetings without affecting Type 2 meetings and CM investment. In case (ii), the baseline economy exhibits overinvestment. A universal CBDC with a small positive interest rate will again improve efficiency in Type 1 meetings and, if it causes the interest rate on deposits to rise, will also improve efficiency in Type 2 meetings and CM investment. Finally, in case (iii), the baseline economy exhibits

Original excerpt: Proposition 7 universal desirability

6.8 Universal CBDC corollaries

The corollaries show two additional implications.

- If project opportunities are sufficiently scarce, a universal CBDC raises welfare.
- If the planner places high value on Type 1 transactions, the optimal universal CBDC interest rate must be high enough to avoid excessive overinvestment.

Interpretation. These corollaries translate the abstract sufficient conditions into policy-relevant statements: universal CBDC is more attractive when the investment opportunity cost is low, and less attractive when the interest rate needed for cash-like benefits would distort deposit-like transactions too much.

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Turning to the question of the optimal interest rate on a universal digital currency, the logic of the first two cases in Proposition 7 can be extended to identify a lower bound. If the policymaker would like to increase production and exchange in Type 1 meetings, then the CBDC interest rate should be large enough to at least ensure there is no overinvestment in the CM.

Corollary 3. If $v > \frac{\alpha\beta}{\mu - (1-\alpha)\beta}$, the optimal interest rate on a universal CBDC satisfies $1+i \geq \frac{\theta\mu}{\beta}$.

In summary, when a central bank cannot target a digital currency to compete only with a single existing type of payment, it must choose the CBDC interest rate to balance multiple concerns. A universal CBDC should not be too attractive relative to cash, to avoid facilitating illicit activities. It should offer an efficient alternative to bank deposits, but not so much as to unduly disintermediate banks. Our analysis shows that, despite the constraints, a CBDC is often desirable. In these cases, the digital currency may only be used in one type of meeting under the optimal policy, even though it is universally accepted. It will tend to compete with bank deposits when productive projects are scarce and/or credit market frictions are strong. In contrast, a universal CBDC will tend to compete with physical currency when productive projects are plentiful and credit market frictions are small.

6. CENTRAL BANK LENDING

7 Central bank lending and synthetic CBDC

7.1 Central bank lending to banks

The paper studies whether central bank lending to banks can neutralize the disintermediation effect of CBDC. The idea is that if households move out of deposits and into CBDC, the central bank could lend the funds back to banks.

7.2 Proposition 8 and equivalence logic

Proposition 8 shows that if the central bank lends an amount b in the deposit market while a deposit-like CBDC is present, Type 2 buyers substitute b units of digital currency for bank deposits. This can make CBDC held in equilibrium without changing the real allocation.

Interpretation. This resembles Brunnermeier–Niepelt equivalence between public and private money: replacing deposits with CBDC backed by loans to banks can leave allocations unchanged.

7.3 Why lending does not solve the main policy problem

The equivalence result holds when the CBDC interest rate is set to replicate the no-CBDC deposit return. But the paper’s earlier analysis shows that this is often not the welfare-maximizing interest rate. The welfare-improving CBDC policy may intentionally raise total outside money balances and reduce the liquidity premium, which necessarily changes bank funding and investment.

Interpretation. Central bank lending can offset mechanical balance-sheet shrinkage, but it cannot remove the fundamental tradeoff between improving payment efficiency and changing the investment margin.

could introduce a digital currency that is held in equilibrium without changing the equilibrium allocation of resources, in line with the Brunnermeier–Niepelt equivalence result.¹⁹

However, our analysis in the previous sections shows that setting $(1+i) = \mu(1+r^N)$ is often not the optimal policy. Instead, Proposition 5 and Figure 3 show that the policymaker can often raise welfare by setting the interest rate higher. Proposition 8 demonstrates that central bank lending cannot mitigate the resulting tradeoff between efficiency in exchange and CM investment because it substitutes one form of inside money (bank deposits) with another (a CBDC backed by loans to the private sector).²⁰ The fundamental tradeoff in our model arises when the policymaker sets the CBDC interest rate to increase the real stock of *outside* money, which promotes efficient exchange but tends to crowd out inside money and decrease investment.

7. CONCLUDING REMARKS

The introduction of a CBDC would represent a potentially historic innovation in monetary policy. If households and firms choose to hold and use significant quantities of such a currency, it could lead to substantial shifts in the types of assets used in exchange as well as in liquidity premia.

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between promoting efficient exchange and efficient investment. If the central bank is able to design separate, targeted digital currencies for each of these uses, it can set the interest rates and other design features to manage each tradeoff. If a digital currency is instead universal, the central bank must design it with both tradeoffs in mind.

Our analysis shows that a cash-like digital currency is desirable if the financial inclusion motive is sufficiently strong. A deposit-like digital currency tends to be desirable when the supply of productive projects is small relative to the transactions demand for deposits and when financial frictions are moderate. If digital currency is universal, these same patterns apply but interact in ways that may lead the currency to circulate either more or less widely than targeted digital currencies would. Taken together, our results show how a digital currency could potentially be an important tool for central banks in managing aggregate liquidity and provide guidance for using this new tool.

The introduction of a CBDC also raises issues that lie outside the scope of our analysis, of course. For example, policymakers have expressed concern that, by providing a safe alternative to bank deposits, a digital currency could facilitate runs on the banking system in periods of financial stress.²¹ Digital currencies can also be held and used internationally much more easily than physical currency, which could potentially alter capital flows and interact with domestic monetary policy. Understanding these issues and how they relate to the fundamental effects of a CBDC identified in our analysis is a promising area of ongoing research.

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Original excerpt: Proposition 8 central bank lending

Original excerpt: concluding policy summary

8 Detailed section-by-section interpretation

8.1 Section 1: Introduction

The introduction motivates CBDC as a response to declining use of physical central bank money and rising interest in digital payments. It also frames the core policy concern: CBDC could pull funds out of bank deposits, reduce stable bank funding, and raise loan rates. The paper’s central contribution is to study this issue in a general equilibrium model where deposits are both bank liabilities and media of exchange.

Interpretation. The introduction sets up the paper against two extreme views: CBDC as purely beneficial payment innovation versus CBDC as pure bank disintermediation risk. The model shows that both are possible depending on design and parameters.

8.2 Section 2: The model

Section 2 builds the environment. Money is essential because DM credit is infeasible. Buyers hold assets before they know what kind of seller they will meet. Sellers can verify only certain assets. Banks issue deposits to finance productive projects, but pledgeability constraints limit deposit supply. The central bank controls outside money through an inflation target and can pay interest on CBDC.

Interpretation. This section creates the mechanism. Verification technology gives assets transactional roles. Pledgeability gives deposits a real investment role. CBDC matters because it changes the return and usability of liquid assets.

8.3 Section 3: No digital currency

Section 3 solves the benchmark equilibrium. Currency supports Type 1 trade, and deposits support Type 2 trade. Bank deposits may carry a liquidity premium. This premium can increase bank-funded investment, which may be good or bad depending on whether investment is below or above the efficient cutoff.

Interpretation. This section is crucial because it shows that bank disintermediation has ambiguous welfare effects. There is no universal presumption that fewer deposits are bad.

8.4 Section 4: Targeted digital currencies

Section 4 studies two targeted CBDCs. A cash-like CBDC competes with currency and affects only Type 1 exchange. A deposit-like CBDC competes with deposits and affects Type 2 exchange plus bank investment. A dual-CBDC system can separately manage the two margins.

Interpretation. Targeted CBDCs are clean theoretical benchmarks. They show exactly what each design does before the paper turns to the realistic constraint that a CBDC may be universal.

8.5 Section 5: Universal digital currency

Section 5 assumes the CBDC is accepted in both types of transactions. The central bank now has only one CBDC interest rate. The key issue is whether that one rate can implement the desired allocation or whether it must compromise between cash-like and deposit-like policy goals.

Interpretation. This section is the heart of CBDC design. It shows that the central bank's problem is not just whether to issue CBDC, but whether CBDC can be made sufficiently targeted in use.

8.6 Section 6: Central bank lending

Section 6 addresses the idea that the central bank could prevent disintermediation by lending CBDC proceeds back to banks. The paper shows that this can preserve allocations under a particular interest rate, but that the welfare-improving CBDC policy often requires changing the liquidity premium itself.

Interpretation. Lending can replicate private money, but it does not replicate the welfare gain from using CBDC as a new outside-money instrument.

8.7 Section 7: Conclusion

The conclusion emphasizes that CBDC can be an important tool for managing aggregate liquidity. The desirability depends on whether the CBDC competes with cash, deposits, or both, and on whether bank-funded investment is too high or too low in the benchmark economy.

Interpretation. The final policy message is conditional and design-focused: central banks should evaluate CBDC through the lens of payment-use margins, deposit liquidity premia, and real investment effects.

9 Short summary

- The paper studies CBDC in a model where both physical currency and bank deposits are used in exchange.
- Deposits can carry a liquidity premium because they serve as a means of payment.
- The liquidity premium lowers banks' funding costs and can increase bank-financed investment.
- A CBDC can improve exchange efficiency by increasing the real balances available for transactions.
- A cash-like CBDC competes with physical currency and does not directly affect bank funding.
- A cash-like CBDC is desirable only if socially valuable cash-type transactions are sufficiently important.
- A deposit-like CBDC competes with bank deposits and can cause disintermediation.
- Disintermediation can be good if the benchmark economy overinvests or if productive projects are scarce.
- Disintermediation can be bad if the liquidity premium on deposits was helping overcome credit frictions.
- A dual targeted-CBDC system can set separate interest rates for cash-like and deposit-like CBDCs.
- A universal CBDC has only one interest rate and therefore must balance multiple margins.
- Universal CBDC can be desirable, but it may circulate more or less widely than targeted CBDCs.
- Central bank lending can make CBDC equivalent to deposits under some policies, but it does not eliminate the fundamental tradeoff when the optimal CBDC policy changes outside-money balances.

10 Conclusion

10.1 Core conclusion

Keister and Sanches show that CBDC design determines welfare. CBDC is not simply good because it improves payments, and not simply bad because it crowds out bank deposits. Its welfare impact depends on the transaction it serves, the value of those transactions, the liquidity premium on deposits, the pledgeability of bank projects, and the scarcity of productive investment opportunities.

10.2 Economic intuition

CBDC changes the composition of liquid assets. If it substitutes for cash, it can improve cash-type exchange without affecting banks. If it substitutes for deposits, it raises the return that private deposits must offer, which can reduce bank funding and investment. Whether that is good depends on whether marginal bank-funded projects are socially valuable.

10.3 Takeaway

The paper's main lesson is that central banks should evaluate CBDC as a tool for managing aggregate liquidity. The key design question is not merely digital versus physical. It is which existing asset the CBDC

competes with and whether the central bank can control that competition through technological restrictions, interest rates, or both.